

1. Administrative & Study Identification

Brief Title: An Open-label Study Comparing Lutetium (177Lu) Vipivotide Tetraxetan (AAA617) in Combination With ARPI Versus AAA617 in PSMA Positive First-line mCRPC **Full Study Title:** A Phase II, Open-label, Multi-Center, Randomized Study Comparing the Combination of Lutetium (177Lu) Vipivotide Tetraxetan (AAA617) and Androgen Receptor Pathway Inhibitor (ARPI) vs. Lutetium (177Lu) Vipivotide Tetraxetan (AAA617) in First-line Treatment of Patients With Prostate-Specific Membrane Antigen (PSMA)-Positive Progressive Metastatic Castration Resistant Prostate Cancer (mCRPC) **Short Title/Acronym:** PSMAndARPI **Protocol Number:** CAAA617B1US01 **NCT Number:** NCT06894511 **Trial Status:** Recruiting **Sponsor Name:** Novartis / Novartis Pharmaceuticals **Investigational Product:** Lutetium (177Lu) vipivotide tetraxetan (AAA617 / Pluvicto) and ARPI (abiraterone [Trade names: Zytiga, Akeega, Yonsa], enzalutamide, apalutamide, or darolutamide) **Phase of Development:** Phase II **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the radiographic progression-free survival (rPFS) response in participants with metastatic CRPC, assessed by conventional imaging, treated with 177Lu-PSMA-617 (AAA617) in combination with an Androgen Receptor Pathway Inhibitor (ARPI) compared to AAA617 alone as a first-line treatment. **Secondary Objectives:** To evaluate overall survival (OS), progression-free survival (PFS), secondary PFS2, prostate-specific antigen (PSA) response, objective response rate (ORR), and the overall safety/toxicity profile.

2.2 Background & Rationale To address a critical gap in the first-line treatment of mCRPC. 177Lu-PSMA-617 binds to PSMA-expressing cells and delivers targeted beta radiation. However, the activation of the androgen receptor (AR) pathway promotes resolution of DNA breaks, leading to radio-resistance. Inhibition of the AR pathway with an ARPI blocks this signaling to promote radio-sensitization and leads to an upregulation of PSMA expression. Combining an ARPI with radioligand therapy is hypothesized to yield synergistic treatment effects and improve clinical outcomes.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (AAA617 alone) **Blinding:** Open-label

Randomization: 1:1, Stratified by type of prior ARPI (abiraterone vs. other [enzalutamide, apalutamide, or darolutamide]) and by the setting of prior ARPI (mHSPC without docetaxel vs. mHSPC with docetaxel vs. others [BCR-non mHSPC or nmCRPC setting]).

3.2 Planned Enrollment & Duration Target Enrollment:

Approximately 420 male participants **Number of Sites:** Multi-center, United States-based study **Estimated Study Start:** 11-Sep-2025 **Estimated Total Duration:** Approximately 3.5 years

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult males ≥ 18 years with an ECOG performance status of 0 to 2. 2. Histopathological and/or cytological confirmation of adenocarcinoma of the prostate. 3. PSMA PET positive disease using FDA-approved PSMA-imaging agents, as determined by the sponsor's central reading rules. 4. Castrate level of serum/plasma testosterone (< 50 ng/dL or < 1.7 nmol/L). 5. Newly diagnosed mCRPC with documented progressive disease after having been previously treated with ARPI in the BCR-non mHSPC, mHSPC, or nmCRPC setting (must have progressed only once on a prior second-generation ARPI: abiraterone, enzalutamide, darolutamide, or apalutamide). First-generation AR inhibitors are allowed but not considered prior ARPI. 6. Must have ≥ 1 metastatic lesion by conventional imaging (CT, MRI, or bone scan) obtained ≤ 28 days prior to randomization. 7. Adequate organ and bone marrow function: * **Bone Marrow:** ANC $\geq 1.5 \times 10^9$ /L, Platelets $\geq 100 \times 10^9$ /L, Hemoglobin ≥ 9 g/dL. * **Hepatic:** Total bilirubin $< 2 \times$ ULN (Gilbert's Syndrome $\leq 3 \times$ ULN permitted); ALT or AST $\leq 3.0 \times$ ULN (or $\leq 5.0 \times$ ULN for liver metastases); Albumin ≥ 2.5 g/dL. * **Renal:** eGFR ≥ 50 mL/min/1.73m² using CKD-EPI equation.

4.2 Exclusion Criteria 1. Previous treatment with radioligand/radiation therapies within 6 weeks of randomization: Strontium-89, Samarium-153, Rhenium-186, Rhenium-188, Radium-223, hemi-body irradiation, and Lu-DOTA radioligand therapy, or any previous PSMA-imaging RLT. 2. Previous treatment with taxane-based chemotherapy in the mCRPC setting. (Taxane exposure is allowed in the mHSPC setting if > 12 months have elapsed since completion). 3. Known or suspected deleterious germline or somatic homologous recombination repair (HRR) gene-mutated mCRPC appropriate for PARP inhibitor therapy. 4. Symptomatic cord compression (or impending findings), or history of CNS metastases that are neurologically unstable, symptomatic, or receiving corticosteroids to maintain neurologic integrity. 5. History of myocardial infarction, angina pectoris, or coronary artery bypass

graft (CABG) within 6 months prior to ICF signature, and/or clinically active significant cardiac disease. 6. Concurrent serious acute or chronic nephropathy as determined by the principal investigator.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - **Arm A:** Lutetium (¹⁷⁷Lu) vipivotide tetraxetan (AAA617) administered at a dose of 7.4 GBq (200 mCi) +/- 10% once every 6 weeks (1 cycle) for a planned 6 cycles, in combination with daily ARPI (abiraterone, enzalutamide, apalutamide, or darolutamide). - **Arm B:** AAA617 alone administered at a dose of 7.4 GBq (200 mCi) +/- 10% once every 6 weeks for a planned 6 cycles. **Route of Administration:** Intravenous (IV) for AAA617; Oral for ARPIs. **Duration of Treatment:** AAA617 for up to 36 weeks (6 cycles); ARPI administration continues until the participant is no longer clinically benefiting or experiences unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Best Supportive Care (BSC) is allowed in both arms. Androgen Deprivation Therapy (ADT) is required to be maintained in both arms. First-generation androgen receptor inhibitor therapy (e.g., bicalutamide) history is allowed. **Prohibited:** Concurrent treatment with PARP inhibitors, taxane-based chemotherapy, or other investigational anti-cancer agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, PSMA PET/CT Scan, Medical History, Labs, Baseline Conventional Imaging (CT/MRI/Bone Scan)
Baseline	Day 0	Randomization, Start of ARPI for Arm A
Interim	Every 6 Weeks	AAA617 Infusions, Safety Labs, Efficacy Assessments (PSA, Imaging), AE monitoring
End of Study	Approx. 3.5 Years	Final safety assessments, survival follow-up, End of Treatment (EoT) visits

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Radiographic Progression-Free Survival (rPFS): Time to radiographic disease progression or death due to any cause. **Measurement Method:** Assessed by Blinded Independent Central Review (BICR) using conventional imaging and PCWG3-modified RECIST v1.1 criteria (soft tissue) and PCWG3 criteria (bone lesions).

7.2 Secondary & Exploratory Endpoints - Overall Survival (OS) - Progression-Free Survival (PFS) - Secondary PFS2 - Objective

Response Rate (ORR) - Incidence and severity of Treatment-Emergent Adverse Events (TEAEs)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via physical examinations, clinical evaluations, and laboratory panels at every treatment cycle. **Serious Adverse Events (SAEs):** Safety events requiring hospitalization, resulting in death, or posing life-threatening risks will be reported per typical 24-hour protocols. **Data Monitoring Committee (DMC):** An Independent Data Monitoring Committee (IDMC) will review efficacy and safety data periodically.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes; Safety Set for all participants receiving at least one dose of study medication. **Sample Size Justification:** Approximately 420 patients randomized 1:1, appropriately powered to detect meaningful shifts in rPFS and OS between the combination arm and monotherapy. **Handling of Missing Data:** Standard survival models (Kaplan-Meier estimates, Cox proportional hazards models) will censor patients accurately for missing time-to-event data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring of clinical data, source documents, and investigational product accountability. **Audit Trail:** Electronic Data Capture (EDC) system with full audit trails, sponsor's central reading rules for PSMA PET eligibility, and query management.

11. Study Identifiers & Governance

Primary ID: CAAA617B1US01 **Registry Number:** NCT06894511 **Posting ID:** 434198530608614606 **Sponsor/Lead Organization:** Novartis / Novartis Pharmaceuticals **Study Title:** PSMAndARPI **Official Regulatory Title:** A Phase II, Open-label, Multi-Center, Randomized Study Comparing the Combination of Lutetium (177Lu) Vipivotide Tetraxetan (AAA617) and Androgen Receptor Pathway Inhibitor (ARPI) vs. Lutetium (177Lu) Vipivotide Tetraxetan (AAA617) in First-line Treatment of Patients With Prostate-Specific Membrane Antigen (PSMA)-Positive Progressive Metastatic Castration Resistant Prostate Cancer (mCRPC)

12. Clinical Framework (Prostate Cancer Specific)

Primary Condition: Metastatic Castration-Resistant Prostate Cancer (mCRPC) **Disease Areas:** Metastatic Castration Resistant Prostate Cancer (mCRPC) **Diagnosis Threshold:** PSMA PET positive disease; castrate testosterone (< 50 ng/dL or < 1.7 nmol/L) **Medical Methodology:** Radioligand Therapy (RLT) combined with ARPI **Optimization Target:** Radiosensitization through AR signaling blockade and upregulation of PSMA expression to maximize DNA-strand breaking

13. Study Procedures & Methodology

Management Pathway: Targeted beta radiation radioligand delivery combined with continuous oral targeted therapy (AR pathway inhibition). **Education Protocol (HCP):** Radiation safety protocols, safe handling/administration of IV AAA617, and combination therapy management. **Education Protocol (Patient):** Radiation precautions post-AAA617 administration, managing potential AEs, and oral adherence to ARPI regimens. **Quality Audit Mechanism:** Central imaging review (BICR) by sponsor-designated readers for PSMA PET eligibility validation and independent radiographic progression verification.

14. Observation & Follow-Up Schedule

Total Duration: Approximately 3.5 years **Onsite Visit Cadence:** Every 6 weeks during the treatment phase (synchronized with the AAA617 dosing cycles). **Remote Monitoring Frequency:** Routine safety follow-ups as determined by specific site protocols post-treatment. **Checklist Review Interval:** Routine imaging assessments every 12 weeks for radiographic progression validation.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				